SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING • HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

EMBEDDED NETWORKING RESEARCH GROUP (ENRG LAB)

Embedded Networking & Autonomous Systems Laboratory • HUST

Pioneering Research in Autonomous Flying Robots (UAVs), Industrial IoT, Edge AI & MR/XR Virtual Cockpits

> *"Pioneering autonomous swarms of flying robots, industrial IoT communications, and real-time artificial intelligence to solve demanding engineering challenges."*

Autonomous Swarms • Low-Power IoT • Edge AI • Digital Twin Solutions

12+	40+	06+
ACTIVE RESEARCHERS	INTERNATIONAL PUBLICATIONS	FUNDED PROJECTS

CORE RESEARCH FOCUS — KEY AREAS

T1 **Digital Twin and Vision Perception-based Autonomous Swarm of Flying robots**

Researching distributed SLAM, vision-based perception, and cooperative control for autonomous flying robot swarms with real-time digital twins.

T2 **MR/XR-based Virtual Cockpit for IoT-connected Unmanned Vehicles**

Developing immersive mixed and extended reality (MR/XR) virtual cockpits for ultra-low latency remote teleoperation over IoT wireless networks.

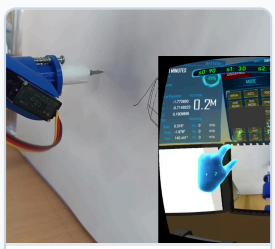
T3 **Adaptive Shared Autonomy for Precise Control of Aerial Manipulators in Contact-Rich Missions**

Pioneering adaptive shared autonomy frameworks uniting human teleoperation and onboard autonomous stabilization for aerial manipulators during physical contact tasks.

T4 **Dynamic Coupling Compensation for Precise Positioning of Aerial Manipulators in Industrial Contact Tasks**

Formulating multi-body dynamic coupling compensation models and dynamic center-of-mass balancing to reject disturbance and ensure high-precision industrial contact operations.

RESEARCH FACILITIES & LIVE DEMOS — DEMOS



Aerial manipulator UAV for high-altitude contact repair



Low-latency MR/XR virtual cockpit for remote teleoperation



Embedded testbench & hardware prototyping at Lab

KEY ACHIEVEMENTS & IMPACT — IMPACT

International Accreditation 2026
Selected as a premier laboratory appraised by international accreditation panels for strategic engineering education and research programs at HUST.

Top Awards in Scientific Research & Innovation
Student research teams won First Prize at the HUST Young Innovators Finals and University-wide Scientific Research Conferences.

National Press & Media Coverage
Featured on Education & Times Newspaper and HUST News for pioneering contact-capable flying robots with real-time digital twins.

Industry Partnerships & Tech Transfer
Conducted specialized engineering training and transferred autonomous inspection and IoT solutions to telecom leaders including MobiFone and Viettel.

STUDENT RESEARCHERS RECRUITMENT

Join ENRG Lab to Elevate Your Research Potential

Member Benefits:

- Direct hands-on experience with UAVs, embedded GPUs (Jetson), HIL testbenches, and advanced dev kits.
- Direct mentorship to author ISI/Scopus papers and compete in prestigious engineering competitions.
- Research stipends, travel funding for competitions, and strong alumni network in the US, Japan, and EU.

Candidate Requirements:

- Undergraduate & graduate students in EE, Telecom, Automation, Computer Science, or Mechatronics.
- Proficiency in at least one language: C/C++, Python, or experience with ROS / ROS2 / MATLAB.
- High responsibility, passion for scientific research, perseverance, and strong self-learning attitude.

Ready to Innovate with ENRG Lab?

Join our dynamic research group to create cutting-edge technology breakthroughs.

Location: Room 715, Building C7, Hanoi University of Science and Technology (HUST)

Lab Director (PI): Dr. Pham Van Tien

Email: tien.phamvan1@hust.edu.vn

Website: tienpv.github.io/web

Fanpage: facebook.com/lab411

Recruitment: Rolling online application & laboratory interview

SCAN QR TO APPLY